

PROJECT 22: MULTIFOCAL DEEP LEARNING MODEL FOR ACCURATE CELL SEGMENTATION IN TRACTION FORCE MICROSCOPY

Benjamin Chambaudet (s240414), Frederik Lundgren (s214626) & Weinan Xiong (214612)

ABSTRACT

This study addresses the challenge of accurate cell segmentation in traction force microscopy (TFM) without relying on fluorescence labeling. The goal is to develop a deep learning model capable of producing precise cell masks using brightfield microscopy images. We implemented and compared two models: a CNN and U-Net using brightfield images from 11 focal depths. The U-Net model achieved the best performance with an IoU score of 63.51% with FFT-filtered images, demonstrating its ability to produce smooth and accurate segmentation masks. With further improvement, our deep learning approach could provide an efficient, non-invasive alternative to traditional fluorescence-based methods for cell segmentation in TFM.

1. INTRODUCTION

Traction force microscopy (TFM) is a powerful technique used to study cellular behavior by measuring the forces exerted by cells on their surroundings. Cells are placed on a grid of nanopillars that bend as cells attach, with their displacement observed via optical microscopy. Accurate segmentation of cells in TFM images is crucial for understanding cellular mechanics. Traditional methods often rely on fluorescence labeling, which can interfere with experimental processes and is not always feasible. To address this limitation, we propose a deep learning-based solution that performs cell segmentation directly on brightfield microscopy images, eliminating the need for fluorescence dyes [11].

Our approach utilizes deep learning models that process 11 images captured at varying focal depths, enabling the model to capture the structure of the cells and their interaction with the nanopillars more effectively. We explore two methods to tackle the segmentation task, U-Net and Convolutional Neural Network, comparing their performance to determine which approach yields the best results. This research presents a step forward in cell segmentation for TFM, offering an efficient, non-invasive alternative to fluorescence-based methods, and has the potential to streamline the analysis of cellular mechanics across a wide range of biological studies.

2. DATA STRUCTURE

The dataset consists of seven wells split into brightfield images and Mask images (Figure 1).

2.1. Brightfield

The Brightfield images are captured at 11 focal planes and serve as the input to the model. These images contain the detailed cell structure across different depths, enabling the model to learn the 3D context of the cells. We have a total of 12.793 brightfield images.

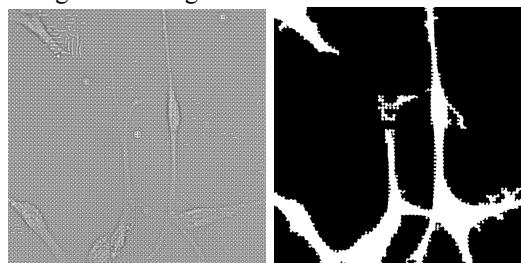


Figure 1, left: Brightfield Image at 1 focal point. right: Mask Image

2.2. Mask

The Mask images are binary representations (one per set of 11 input images) that specify the correct output or cell mask for each corresponding set of Brightfield images. These masks are taken from focal point 6 and are used for training the model to learn accurate cell boundaries. We have 1.163 masks. They represent the correct cell location.

3. PREPROCESSING

Preprocessing refers to the methods applied to prepare the data, ensuring it is clean, structured, and ready for the model to learn effectively. We applied basic standardization along with additional techniques to clean, enhance, and structure the data before training.

3.1. FFT

Fast Fourier Transform (FFT) is a method that can be used on a 2D image to separate the frequencies. With the use of an FFT, we can decompose the image into sinusoidal waves, allowing for the processing of specific frequencies [9]. This is important for us, as the bed the cells are placed on consists of a series of microneedles. In the frequency domain of the FFT, the microneedles will appear at the same frequency, which will allow us to filter these and smoothen them out. Thus we can use FFT as a filter, improve the overall quality of the data visually, and give the model a

better opportunity to extract useful features and create more accurate segmentations.

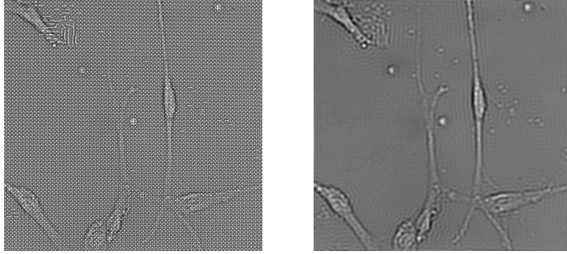


Figure 2, left: Original Image, right: Filtered image using FFT

3.2. Data augmentation

Data augmentation is a method to create variation in a dataset, to either gain more data or vary current data enough to create a more generalized model [6]. We found that enough data was available to produce acceptable segmentations from early pilot runs of our CNN and U-Net models. Because of this, we did not want to expand the data set by augmentation. We instead used horizontal and vertical flips to augment data and create variation, as well as rotation so our model learns to segment the cells from different angles. Each of the augmentation methods had a 50% chance of being applied, with a random variation in the rotation. Because the images come in groups of 11 with 1 matching mask for the group the augmentations are used on the entire group as well as their corresponding mask. This ensures that the mask will always fit the data and that we do not introduce mistakes.

3.3 Train, Validation and Test

Before training the model a split in the data was made. Here well 3-7 will be training data, well 2 will be validation and well 1 will be test data. This split in the data is performed to gain a reliable model with trustworthy metrics. A training set of data will be the data our model learns from. This data will be used to set the weights and biases of the model, so the model knows important features of the data. To evaluate the model we use the validation data. This data is used to fine-tune hyperparameters and get an understanding of the model fit on “unlearned” data. The model will not be trained on the validation set but will be affected indirectly as we use this to improve overall model performance.

To ensure an unbiased evaluation of the final model fit the test data is used. This data has been kept away from the model throughout the entire training and fine-tuning process. It is only as the last step it will be used, as a metric of how well the trained model performs [1].

4. MODELS

Using our dataset, which is composed of the brightfield image together with the provided mask, we can make a machine-learning model that can segment an arbitrary brightfield image/images to find the cell. To achieve the best

result, we conducted tests with two different models in this study: CNN and U-Net.

4.1. Convolutional Neural Network

The Convolutional Neural Network (CNN) model is designed to process 11-channel brightfield microscopy images for cell segmentation, employing an encoder-decoder architecture to extract and reconstruct intricate spatial features. The encoder compresses the input image into a lower-dimensional feature representation through three convolutional layers, each followed by a ReLU activation and a max pooling operation. These convolutional layers progressively increase the number of filters from 64 to 128 and then to 256, allowing the model to learn increasingly abstract features. Max pooling operations reduce the spatial dimensions by half after each layer, capturing larger spatial contexts while reducing computational costs [13].

The decoder reconstructs the feature representation back to the original image size using three transposed convolutional layers. These layers upsample the feature maps, with the number of filters decreasing from 256 to 64, before producing the final output with a single filter. Each transposed convolution is followed by a ReLU activation, except for the final layer, which uses a sigmoid activation function to output pixel values between 0 and 1. This sigmoid output is appropriate for binary segmentation, where each pixel is classified as belonging to either the foreground or background [12].

This architecture is particularly suitable for binary segmentation tasks due to its ability to extract hierarchical features during encoding and restore spatial resolution during decoding. The encoder captures low-level to high-level features, while the decoder ensures that these features are mapped back to the original image size. The simplicity of this model allows for efficient training and interpretability, making it effective for segmentation tasks involving large images, such as those in medical imaging or microscopy [14].

4.2. U-Net

U-Net is a well-known model for medical image segmentation. It is a fully convolutional neural network that specializes in image segmentation [7]. We use MONAI (Medical Open Network for AI) prebuilt neural network architecture [8]: BasicUNet for the implementation. U-Net can be used to determine the presence of objects within an image and generate a corresponding mask indicating the object’s location. The architecture is divided into two parts just like the cnn: the encoder and the decoder. The encoder breaks the image into smaller pieces to find important features and the decoder rebuilds the image while focusing on the features that matter most. These two parts are connected using skip connections so that details from the

encoder are reused in the decoder. This helps the model keep important information [10].

The U-Net model takes 11 brightfield images as input and produces a segmentation map for each class as output. This map represents the model's prediction of the input image's segmentation. The number of feature channels at each level of the U-Net used is (32, 64, 128, 256, 512, 32). The first five values correspond to the encoder feature sizes at each of the five levels. The last value corresponds to the feature size after the last upsampling. The activation function used is Leaky ReLU (Rectified Linear Unit). Leaky ReLU allows a small, positive gradient for negative input values that prevents these neurons from the "dying ReLU" problem during training. Softmax is used on the results after the model finishes analyzing the image. It turns the raw numbers into probabilities showing how likely each pixel in the segmentation map is to belong to each category.

4.3. Intersection over Union

We need a way to evaluate the performance of the models. In image segmentation tasks, Intersection over Union (IoU) is a common evaluation metric used to measure the overlap between the predicted segmentation output and the ground truth labels. IoU quantifies the accuracy of the model's predictions by computing the ratio of the intersection area to the union area of the two sets [5].

4.4. Model optimization

After testing various configurations, we found that the following hyperparameters yielded the best results for our model:

- batch size = 16
- Learning rate = 0.001
- Number of epochs = 30

To optimize performance, we experimented with two widely used loss functions: Generalized Dice Loss (GDL) and Binary Cross-Entropy (BCE) Loss. These loss functions are particularly effective for image segmentation tasks, as they help the model focus on minimizing segmentation errors and improving overall accuracy [2][4].

5. EXPERIMENTS AND RESULTS

We aim to create a model that could accurately segment cells in 2D brightfield images, to achieve this we experimented with different variations of CNN and U-Net. We evaluated both models and compared their performance using a combination of quantitative and qualitative methods. First, we tried to see the basic performance of the models using the hyperparameters described earlier and giving all 11 images to predict the mask. To measure the performances we used metrics Intersection over Union.

5.1. Convolutional Neural Network

We evaluated the performance of our convolutional neural network (CNN) using two loss functions, Binary

Cross-Entropy (BCE) Loss and Generalized Dice Loss (GDL), and compared results with and without FFT filtering for noise reduction. The models were assessed based on the Intersection over Union (IoU) metric.

In the first experiment, we trained the CNN and used the GDL loss function. This model achieved an average IoU of 40.49%. The segmentation masks produced in this configuration were noisy, with numerous false positives and poorly defined cell boundaries. The model struggled to capture fine details, likely due to high-frequency noise which can significantly affect the loss calculation for GDL. Indeed, when noise is present in the background and/or foreground, the weighted contributions can be altered causing the GDL to misinterpret noisy regions as significant foreground. This resulted in lower segmentation accuracy.

In the second experiment, we trained the CNN but switched to the BCE loss function. This model achieved a slightly higher average IoU of 42.05%. The BCE loss function improved segmentation performance compared to GDL by producing masks with better-defined structures and fewer false positives. However, high-frequency noise in the brightfield images continued to limit the model's accuracy, preventing further improvements.

In the final experiment, we applied FFT filtering to remove high-frequency noise and used the BCE loss function. This configuration significantly improved the segmentation performance, achieving an average IoU of 53.09%. The predicted masks better resembled the ground truth, with clearer cell boundaries and minimal noise. This demonstrates the effectiveness of FFT filtering in enhancing segmentation accuracy by eliminating noise that hinders model performance. Despite the noise reduction, the validation loss exhibits significant fluctuations, characterized by high variance and sudden spikes, possibly due to overfitting or residual noise in the brightfield images. To mitigate this inconsistency, incorporating additional regularization techniques and experimenting with different learning rates could help improve model stability and generalization performance. (see Appendix 1)



Figure 3, left: CNN with GDL, middle: CNN with BCE, right: CNN with BCE and FFT

The visual comparison highlights that combining FFT filtering and BCE loss yields the most accurate segmentation, with significantly reduced noise and better delineation of cell structures. These findings underscore the importance of selecting the appropriate loss function and applying noise reduction techniques to improve CNN-based segmentation performance.

5.2. U-Net

For the U-Net the generalized dice loss (GDL) was used. We see the train and validation error decrease steadily over the epochs indicating the model continues to learn as shown in Appendix 2. The train validation and test time used in total was 5498.43 seconds when using all the images. U-Net demonstrated stronger performance on the brightfield images with an average IoU of 63.38% on the test well. Evaluating the IoU metric for each image closely we see some variation in the performance.



Figure 4, left: U-Net, middle: U-Net with FFT, right: U-Net with 1 input

Figure 4 left shows that U-Net produces segmentation masks with smooth boundaries in most cases. U-Net is effective in accurately segmenting contiguous cells with moderate complexity. For example, in regions with clear contrast between object and background, it captures object boundaries effectively. However, in challenging scenarios such as objects with fine details, blurry boundaries, or unusual shapes, the model occasionally misclassifies pixels, leading to either over-segmentation or under-segmentation. (Examples shown in Appendix 3).

An approach using FFT to remove high frequency was also used on the U-Net, where we precomputed the images with an FFT high band filter and used the filtered images as input. The IoU was very similar, with a slight increase, being 63.51%. The FFT-filtered input images performed similarly to the original images when used in a U-Net, but the segmentations did appear to have a smoother edge using FFT input.

To make the U-Net more computationally efficient we reduced the number of input images. The provided masks are only from focal point 6, so we assumed that the sixth layer would contain enough information. Using just the sixth layer as input gave a decreased IoU of 59.14%, but a significantly reduced runtime of 1539.93 seconds.

6. DISCUSSION

Comparing different models such as our CNN and our U-Net will typically be difficult due to the underlying differences in the models. We tried to keep as many variables the same and used an encoder-decoder architecture for both models to ensure a fair comparison. We found that, with its skip connections, the U-Net performed significantly better, which is to be expected since the U-Net is specifically designed for segmentation tasks. The U-Net achieved an IoU of 63.51%, while CNN's best performance was an IoU of just 53.09%. This demonstrates that the

U-Net is the preferred model for this task, as it is better suited to capturing both low-level and high-level features necessary for accurate segmentation.

The use of FFT filtering in preprocessing was explored as a method to improve the quality of brightfield images by removing high-frequency noise caused by the microneedle bed. Our results showed that applying FFT filtering slightly improved segmentation accuracy for both the CNN and U-Net models, as evidenced by an increase in IoU scores. While the improvement was relatively small for the U-Net, the segmentation masks appeared visually smoother, particularly at the cell boundaries.

A key consideration in our study was the trade-off between accuracy and computational efficiency. While the multifocal approach using all 11 focal planes with FFT filtering provided the best segmentation accuracy, it came at the cost of significantly longer training and inference times. In comparison, reducing the input to a single focal plane decreased the runtime to 1539.93 seconds a threefold improvement (not counting the FFT preprocessing time) from using all 11 layers at the expense of accuracy, where the IoU dropped to 59.14%.

Future work could focus on optimizing models through layer analysis or lightweight architectures to improve computational efficiency without sacrificing segmentation accuracy. Additionally, integrating skip connections into the CNN model may help determine whether these connections are responsible for the U-Net's superior performance.

7. CONCLUSION

This study presented a deep learning-based approach for accurate cell segmentation in TFM using brightfield images, eliminating the need for fluorescence labeling. Using multifocal input across 11 focal planes, we compared two models: a Convolutional Neural Network (CNN) and a U-Net architecture to evaluate their segmentation performance. Our results demonstrated that U-Net outperformed the CNN model, achieving an IoU score of 63.38% on the test set while maintaining smooth segmentation boundaries. The CNN model, though computationally lighter, delivered much lower accuracy even when FFT filtering was added. To balance efficiency and accuracy, we tested a reduced input approach using only the 6th focal plane, which achieved an IoU of 59.14% with a significant decrease in runtime.

The findings highlight the strength of U-Net, in handling complex segmentation tasks directly from brightfield microscopy images which offers a non-invasive, efficient, and scalable alternative to traditional methods. With further improvements, this deep learning model could make TFM analysis faster and more efficient, while providing accurate cell segmentation for many types of biological research.

8. REFERENCES

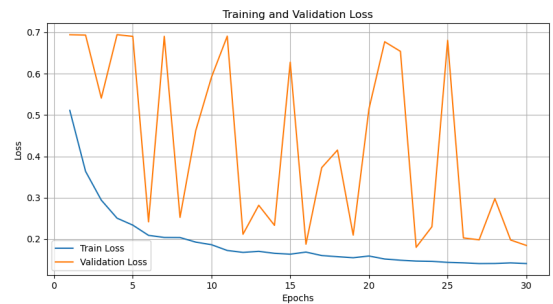
- [1] Shah, T. (2021, May 24). About train, validation and test sets in machine learning. Medium.
<https://towardsdatascience.com/train-validation-and-test-sets-72cb40cba9e7>
- [2] PyTorch Contributors (2023).
<https://pytorch.org/docs/stable/generated/torch.nn.BCELoss.html>
- [3] Jadon, S. (2020, October). A survey of loss functions for semantic segmentation. In 2020 IEEE conference on computational intelligence in bioinformatics and computational biology (CIBCB) (pp. 1-7). IEEE.
- [4] Sudre, C. H., Li, W., Vercauteren, T., Ourselin, S., & Jorge Cardoso, M. (2017). Generalised dice overlap as a deep learning loss function for highly unbalanced segmentations. In Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support: Third International Workshop, DLMIA 2017, and 7th International Workshop, ML-CDS 2017, Held in Conjunction with MICCAI 2017, Québec City, QC, Canada, September 14, Proceedings 3 (pp. 240-248). Springer International Publishing.
- [5] Rosebrock, A. (2024, November 18). Intersection over Union (IoU) for object detection - PyImageSearch. PyImageSearch.
<https://pyimagesearch.com/2016/11/07/intersection-over-union-iou-for-object-detection/>
- [6] Awan, A. A. (2024, December 9). A Complete Guide to Data Augmentation. Datacamp.
<https://www.datacamp.com/tutorial/complete-guide-data-augmentation>
- [7] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-net: Convolutional networks for biomedical image segmentation. In Medical image computing and computer-assisted intervention—MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18 (pp. 234-241). Springer International Publishing.
- [8] Cardoso, M. J., Li, W., Brown, R., Ma, N., Kerfoot, E., Wang, Y., ... & Feng, A. (2022). Monai: An open-source framework for deep learning in healthcare. arXiv preprint arXiv:2211.02701.
- [9] R. Fisher, S. Perkins, A. Walker and E. Wolfart. (2003) Image Transforms - Fourier Transform.
<https://homepages.inf.ed.ac.uk/rbf/HIPR2/fourier.htm>
- [10] GeeksforGeeks. (2023, June 8). UNET Architecture explained. GeeksforGeeks.
<https://www.geeksforgeeks.org/u-net-architecture-explained/>
- [11] Tan JL, Tien J, Pirone DM, Gray DS, Bhadriraju K, Chen CS. Cells lying on a bed of microneedles: An approach to isolate mechanical force. Proceedings of the National Academy of Sciences 2003;100(4):1484–1489.
- [12] Prosise, J. (2023, December 16). Binary Classification with Neural Networks - Atmosera. Atmosera.

<https://www.atmosera.com/blog/binary-classification-with-neural-networks/>

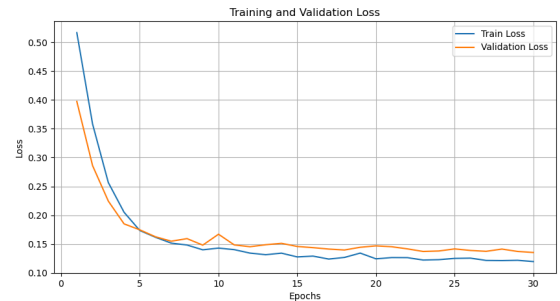
- [13] Zhao, L., Zhang, Z. (2024, January 18). An improved pooling method for convolutional neural networks. Sci Rep 14, 1589.
<https://doi.org/10.1038/s41598-024-51258-6>

- [14] Sihang Zhou, Dong Nie, Ehsan Adeli, Jianping Yin, Jun Lian, Dinggang Shen (2019, June 19). High-Resolution Encoder–Decoder Networks for Low-Contrast Medical Image Segmentation, National Library of Medicine.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8195630/>

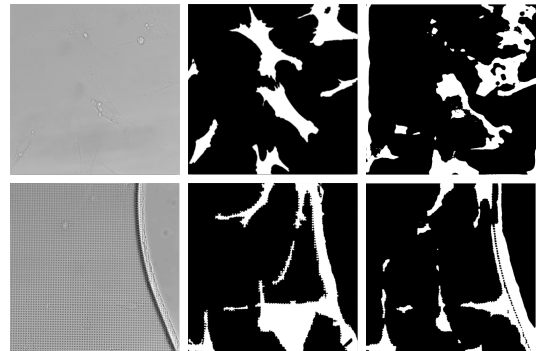
9. APPENDIX



Appendix 1, Train-Validation plot for the CNN model using BCE loss



Appendix 2, Train-Validation plot for the U-Net model



Appendix 3, Examples of misclassifications from U-Net
left: Brightfield images, middle: ground truths, right: predictions

10. GITHUB REPOSITORY

<https://github.com/WeinanXiong/P22-Multiview-segmentation.git>